Netflix Titles Dataset: Data Cleaning, Exploratory Analysis & Business Insights

Objective

The objective of this analysis was to clean, structure, and explore the Netflix Titles dataset using Data Analysis tools, SQL and Excel uncover meaningful content trends, improve data quality, and generate business-relevant insights from the dataset.

Dataset Overview

Metric	Value
Total Rows	8,808
Total Columns	12
Missing Values	4,307
Primary Key	Show ID
Main Categories	Movies, TV Shows

Data Cleaning Process

The following data cleaning steps were performed in SQL Server:

- Imported raw Excel dataset in CSV format into SQL Server
- Corrected inconsistent data types
- Standardized text formatting using uppercase and trimming functions
- Converted date fields into proper DATE format
- Replaced missing values in:

    - Director
    - Cast
    - Country
    - Rating
    - Duration
    - Date
- Removed duplicate records using ROW_NUMBER()
- Validated release years and checked for invalid entries
- Exported Cleaned dataset back into Excel
- Extracted the Year from Date_Added using Power Query.

Exploratory Data Analysis (EDA)

The dataset was analyzed to identify:

- Distribution of Movies vs TV Shows
- Top 5 most common content ratings
- Top 10 countries producing Netflix content
- Content growth trends over time
- Relationships between release year and ratings

Visualizations Included

- Bar Chart — Top Producing Countries
- Histogram — Release Year Distribution

- Time-Series Plot — Content Added by Year
- Column Chart – Most Common Ratings
- Pie Chart – Distribution of Movies vs TV shows

Key Insights

- Movies dominate the platform, representing the majority of Netflix content.
- The United States contributes the highest number of titles on Netflix.
- TV-MA is one of the most frequent ratings, indicating strong mature-audience targeting.

Business Implications

- Netflix prioritizes movie content over TV shows.
- Growth in recent content additions suggests aggressive platform expansion.
- Country-level analysis can support regional content investment decisions.

Conclusion

The Netflix dataset was successfully cleaned and transformed into a structured format suitable for analysis. The exploratory analysis revealed important trends in content distribution, ratings, and regional production, providing insights that can support strategic content and audience decisions.

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